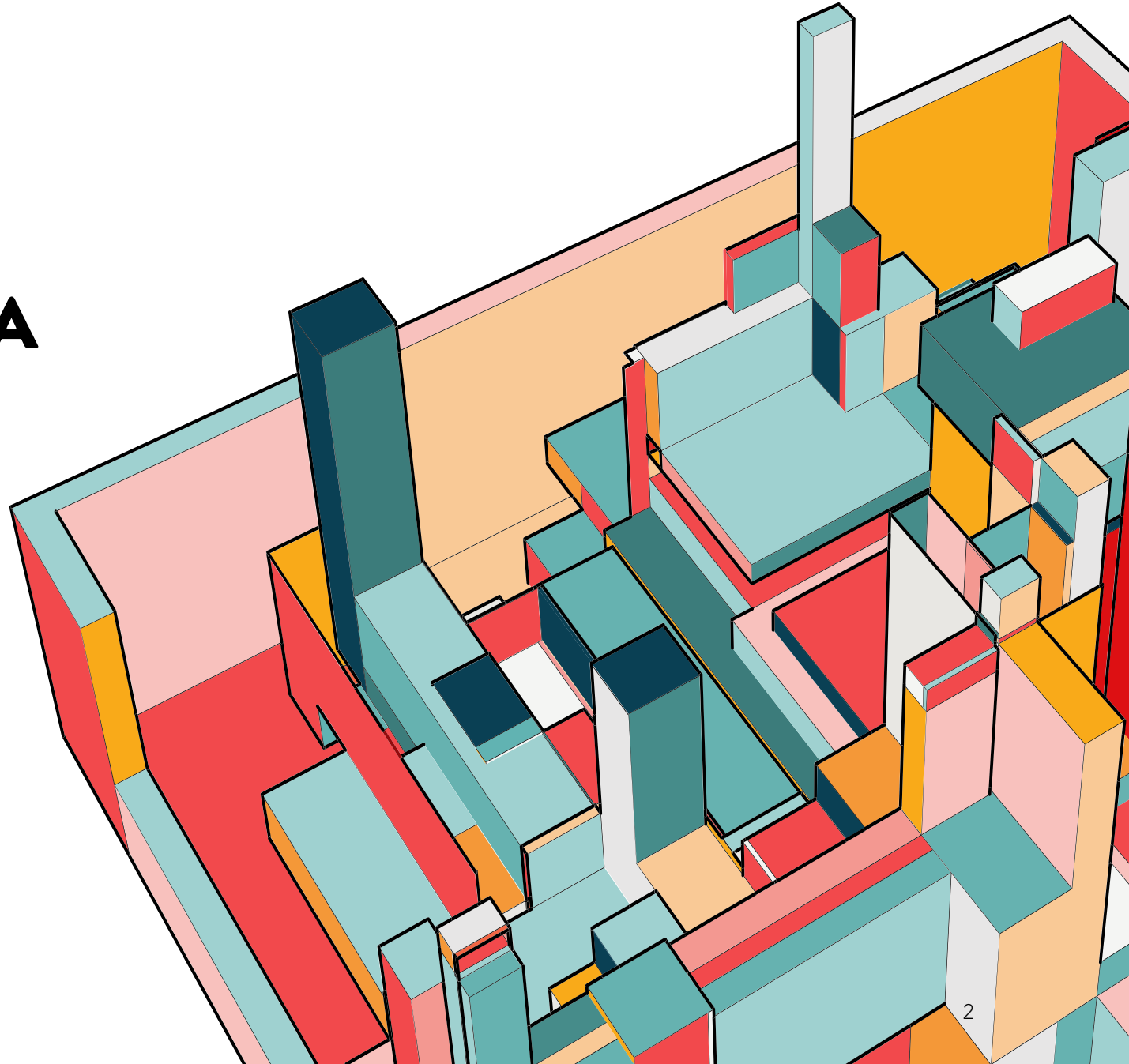
DATA AND INFORMATION IN A FAST-CHANGING ERA

Data and Information, are they the same?

What is happening with data/information in this era?



DATA VISUALIZATION – WHY WE NEED IT?

COMMUNICATING

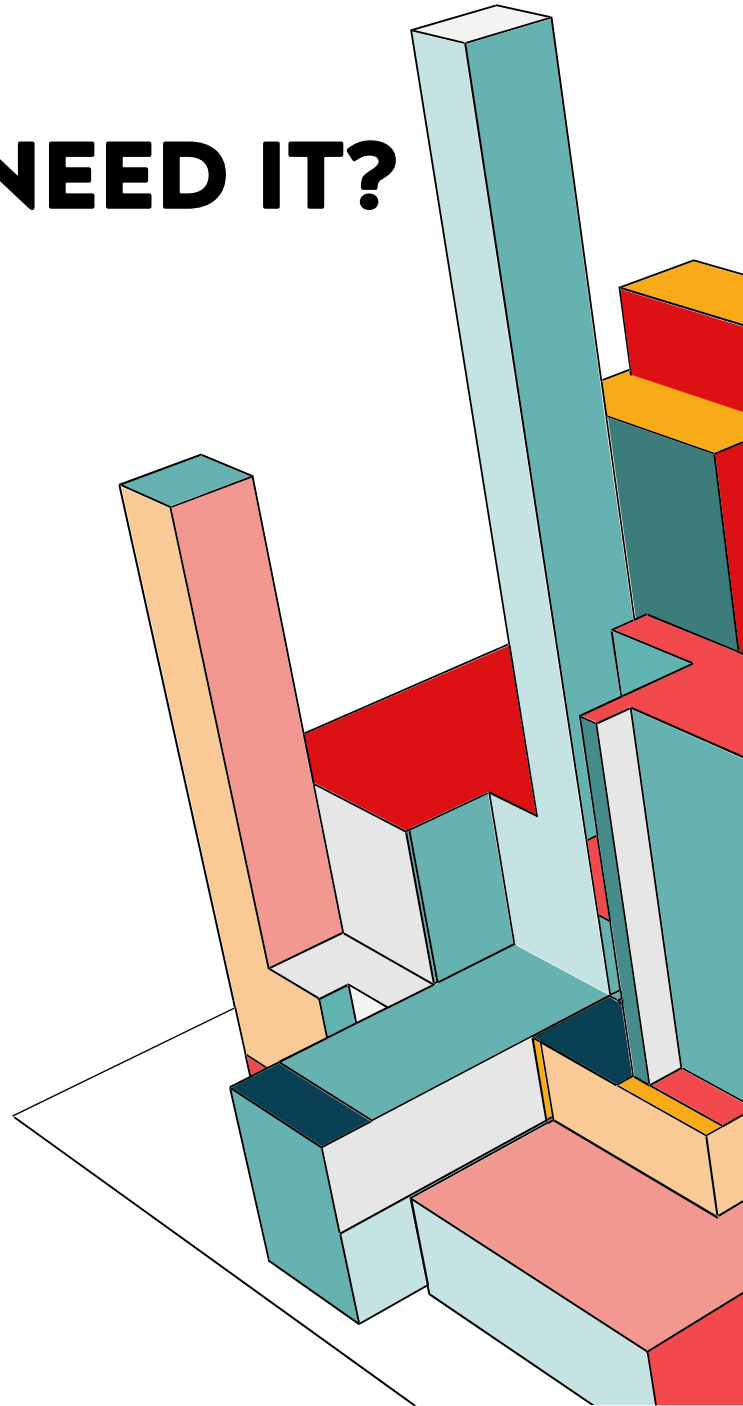
For hundreds of years, we used visuals to communicate/deliver information. And that hasn't changed.

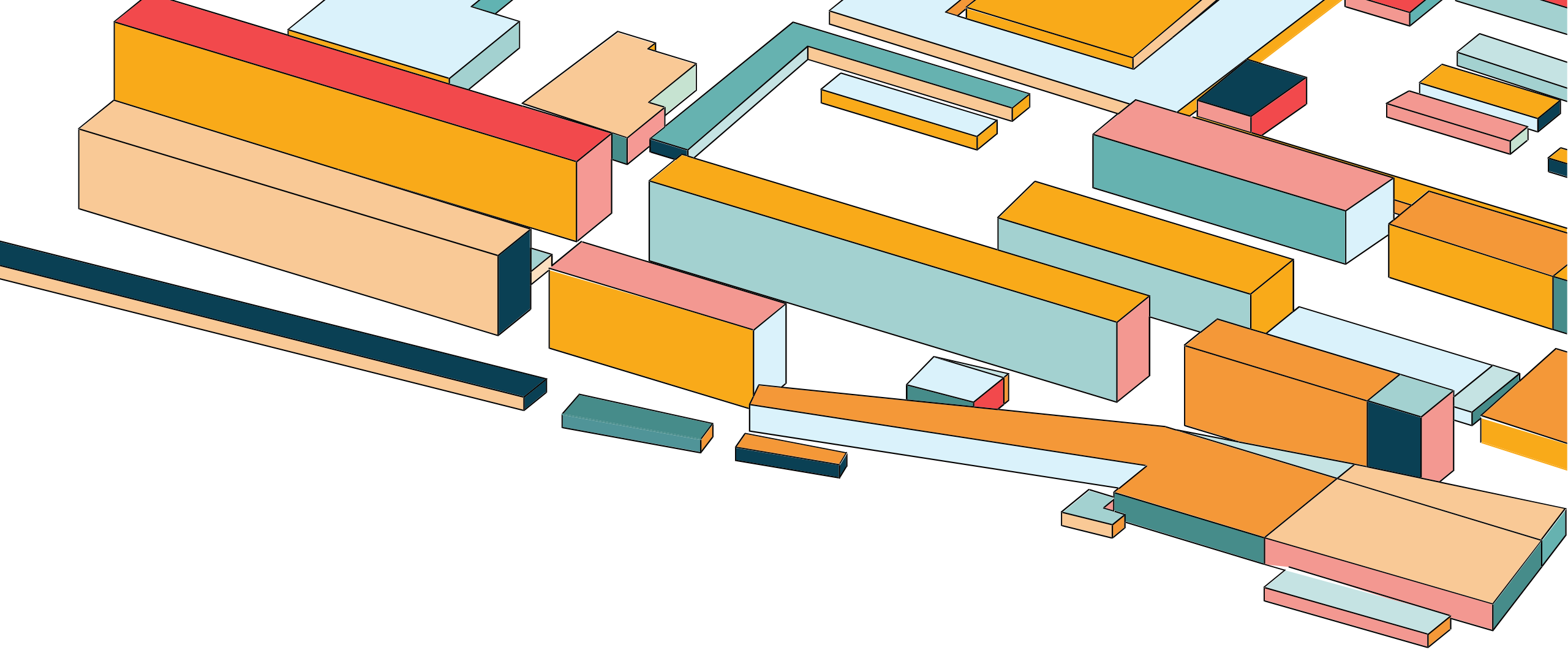
HANDLING INFORMATION DENSITY

The amount of data/information increase and the way we consume information has differed rapidly

REVEALING YOUR DATA

The very first step of analysis



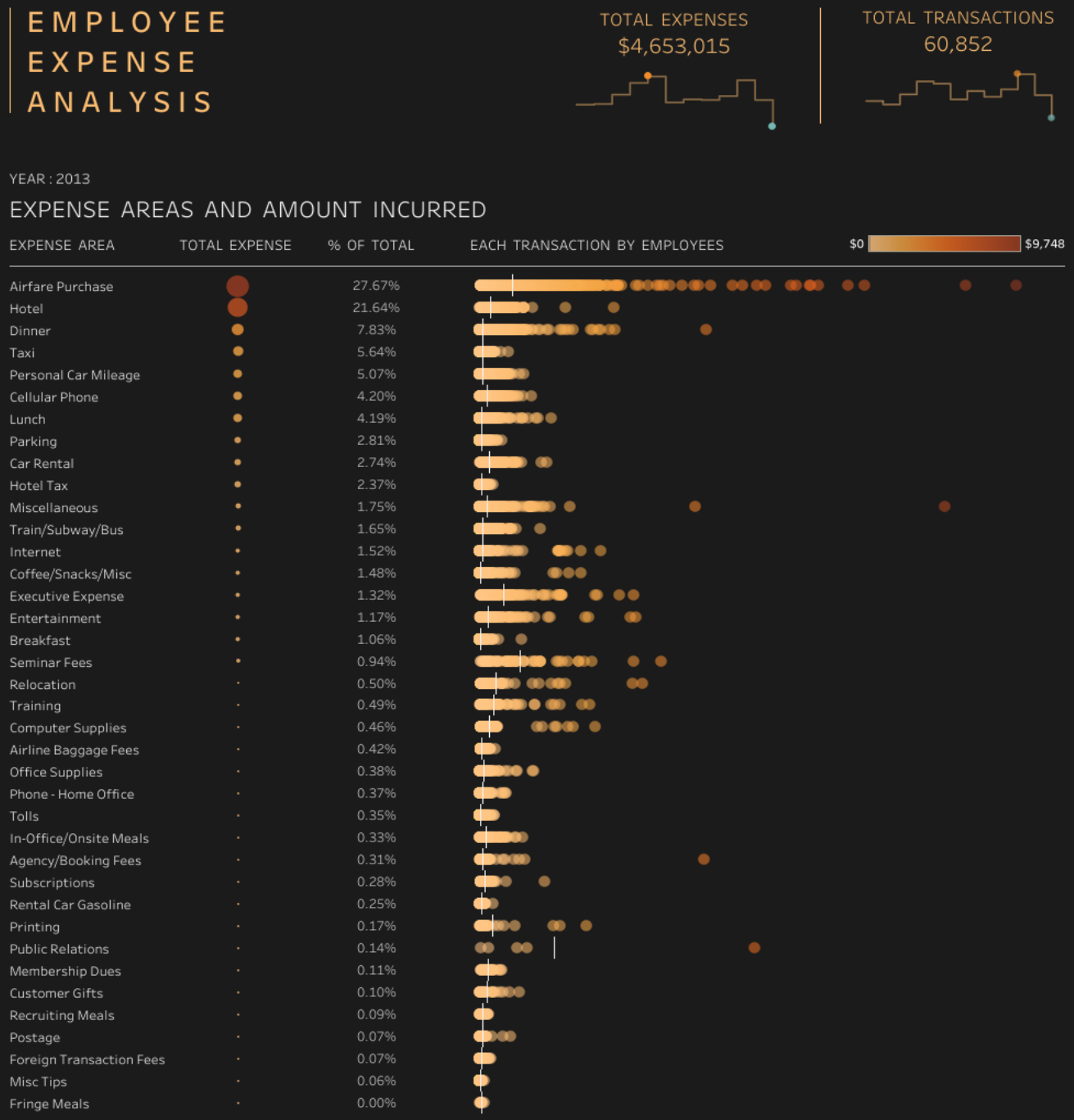


A GOOD VISUALIZATION?

TECHNIQUES

- ✓ Customized plot chart with quantitative and nominal variables
- ✓ Highlighting
- ✓ Interactivity
- ✓ Informative titles
- ✓ Avoid using Pie chart





FRAMEWORKS APPLIED

BERTIN'S SEMIOLOGY

- Position – Ordered, Quantitative
- Value – Ordered, Quantitative
- Size – Ordered, Quantitative

MACKINLAY'S RANKING

- Quantitative – Position, Density (Value), Area (Size)
- Nominal - Position

THE FINANCIAL TIMES VISUAL VOCABULARY

- Magnitude
- Ranking

GESTALT PRINCIPLES

- Similarity
- Focal point

WHAT MAKES IT GOOD?

BEAUTIFUL

Contrast colours

INFORMATIVE

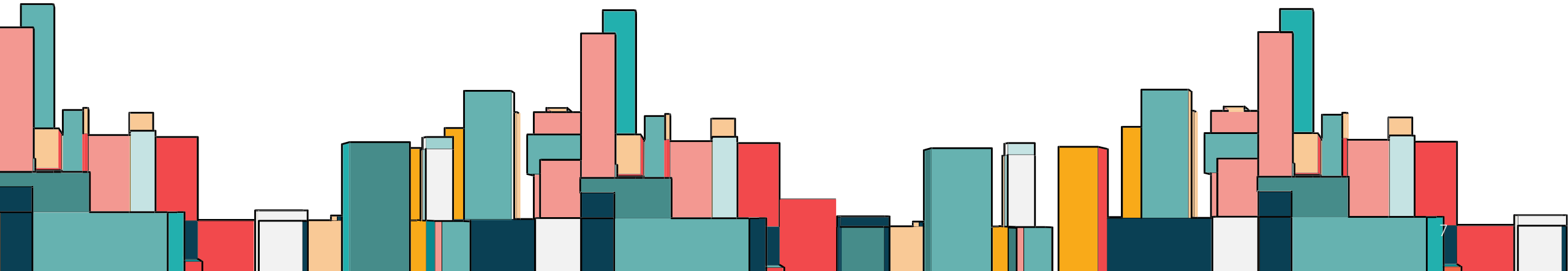
Deliver information
Intended message

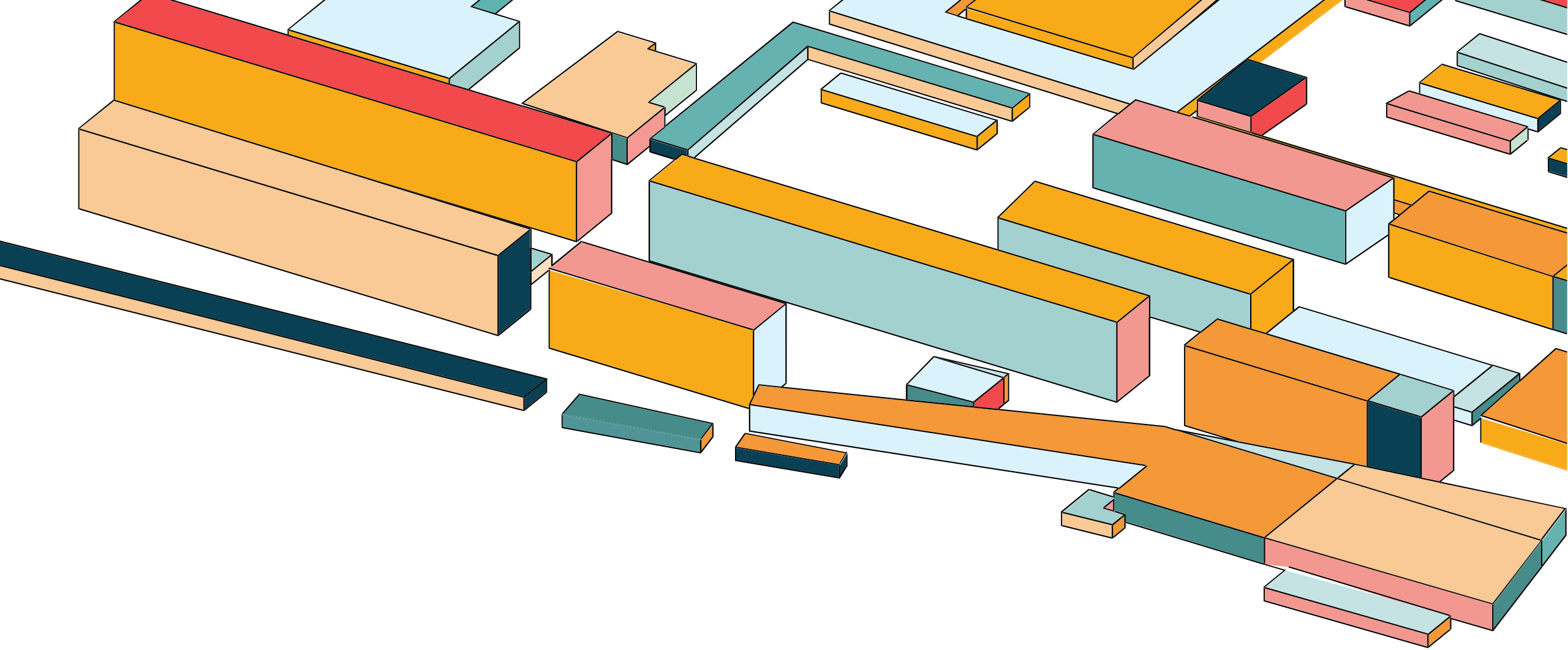
EFFICIENT

Clear
Simple layout

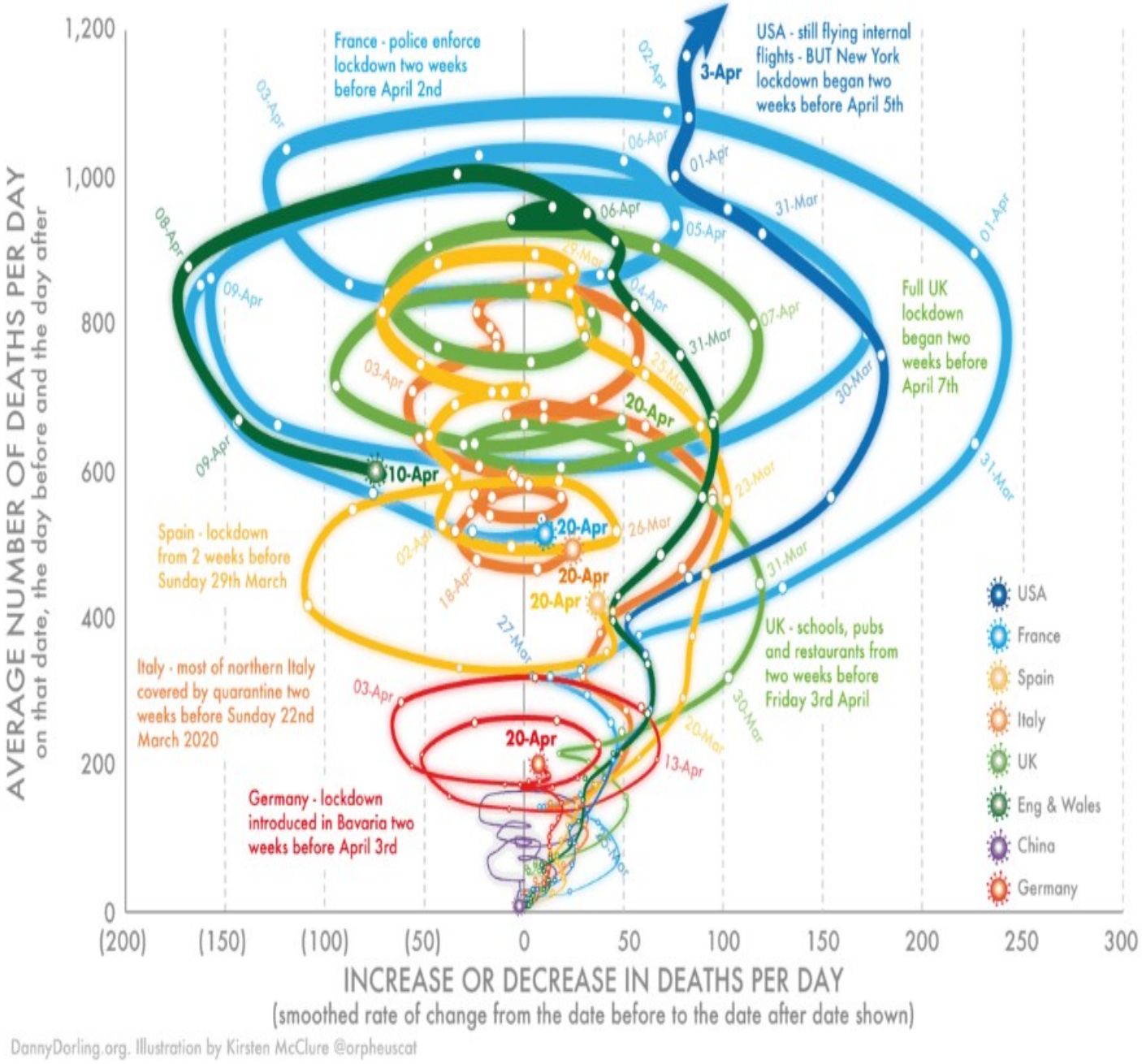
NOVEL

Detailed information on “Expense Area” and “Each transaction by employees”





A BAD VISUALIZATION?



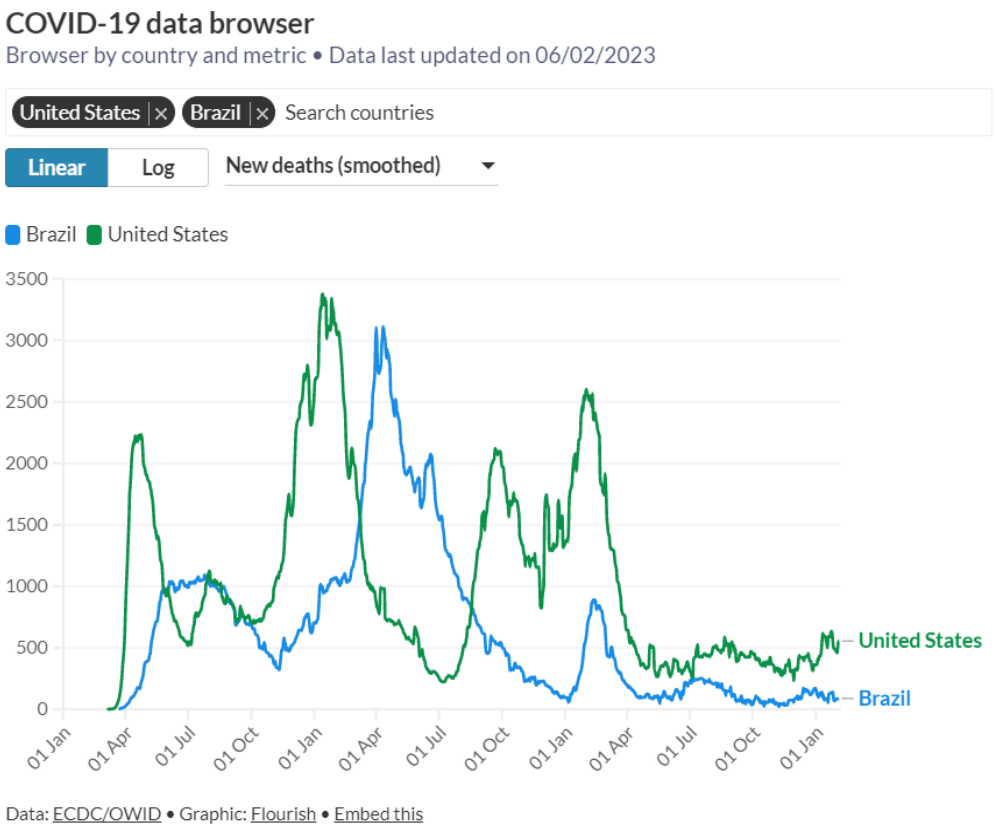
PROBLEMS

- X Too many annotations
- X Inappropriate way to present time-series data
- X Colors and dots make it much more cluttered
- X Confusing message

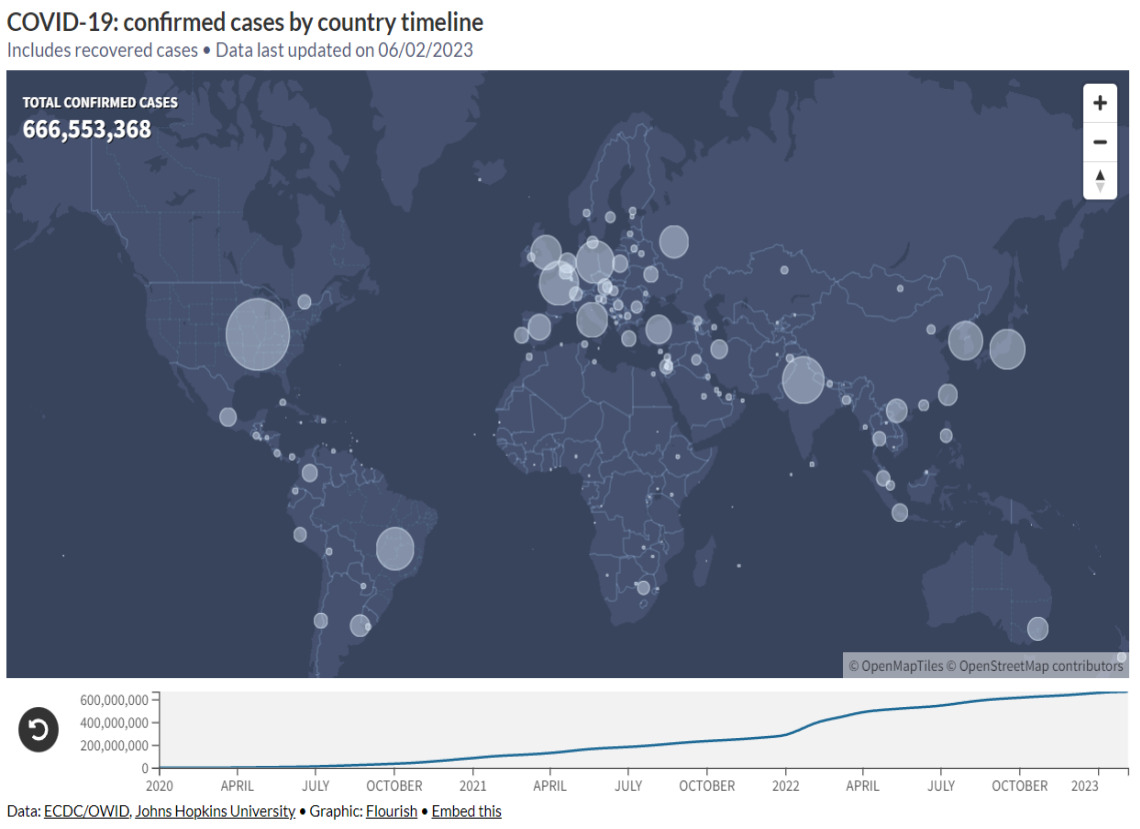
RECOMMENDATION

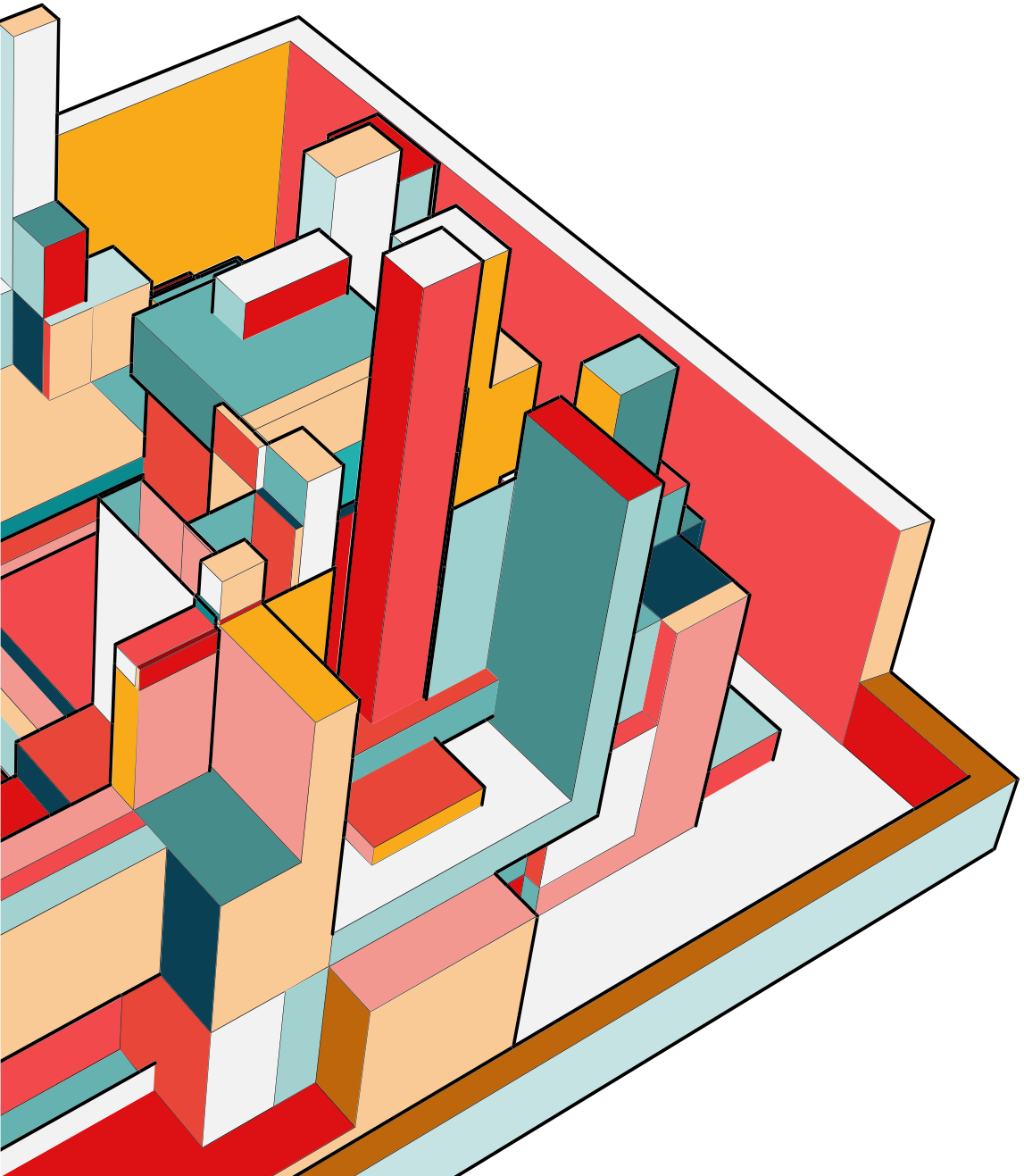
- For the bad visualization -

LINE GRAPH



GEO-MAP WITH PROPORTIONAL SYMBOLS





SUMMARY

- The objective of any graphic in the context of presentations is to effectively convey information.
- Creating visualizations is a process of encoding information into signs and make the decoding as easy as possible for the receiver.
- Understanding the dataset and revealing the key/hidden information by picking up significant visualization types

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Mackinlay, J. (1986). Automating the design of graphical presentations of relational information. ACM Transactions on Graphics, 5(2), 110–141. https://doi.org/10.1145/22949.22950

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